Research Analysis Report

Research Topic: Retrieval Augmented Generation

Total Papers: 5

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Executive Summary

This report provides a technical overview of 'Retrieval Augmented Generation' including methodologies, research gaps and experiment planning.

Literature Review

Introduction: This literature review summarizes findings from 5 research papers related to the selected topic. It highlights research trends, methodologies, key contributions, future directions, and overall observations.

Paper Summaries

• {'title': 'Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks', 'year': 2020, 'summary': 'Large pre-trained language models have been shown to store factual knowledge in their parameters, and achieve state-of-the-art results when fine-tuned on downstream NLP tasks. However, their ability to access and precisely manipulate knowledge is still limited, and hence on knowledge-intensive tasks, their performance lags behind task-specific architectures. Additionally, providing provenance for their decisions and updating their world knowledge remain open research problems. Pre-trained models with a differentiable access mechanism to explicit non-parametric memory can overcome this issue, but have so far been only investigated for extractive downstream tasks. We explore a general-purpose fine-tuning recipe for retrieval-augmented generation (RAG) -- models which combine pre-trained parametric and non-parametric memory for language generation. We introduce RAG models where the parametric memory is a pre-trained seq2seq model and the non-parametric memory is a dense vector index of Wikipedia, accessed with a pre-trained neural retriever. We compare two RAG formulations, one which conditions on the same retrieved passages across the whole generated sequence, the other can use different passages per token. We fine-tune and evaluate our models on a wide range of knowledge-intensive NLP tasks and set the state-of-the-art on three open domain QA tasks, outperforming parametric seq2seq models and task-specific retrieve-and-extract architectures. For language'}

• {'title': 'Retrieval-Augmented Generation for Large Language Models: A Survey', 'year': 2023, 'summary': "Large Language Models (LLMs) showcase impressive capabilities but encounter challenges like hallucination, outdated knowledge, and non-transparent, untraceable reasoning processes. Retrieval-Augmented Generation (RAG) has emerged as a promising solution by incorporating knowledge from external databases. This enhances the accuracy and credibility of the generation, particularly for knowledge-intensive tasks, and allows for continuous knowledge updates and integration of domain-specific information. RAG synergistically merges LLMs' intrinsic knowledge with the vast, dynamic repositories of external databases. This comprehensive review paper offers a detailed examination of the progression of RAG paradigms, encompassing the Naive RAG, the Advanced RAG, and the Modular RAG. It meticulously scrutinizes the tripartite foundation of RAG frameworks, which includes the retrieval, the generation and the augmentation techniques. The paper highlights the state-of-the-art technologies embedded in each of these critical components, providing a profound understanding of the advancements in RAG systems. Furthermore, this paper introduces up-to-date evaluation framework and benchmark. At the end, this article delineates the challenges currently faced and points out prospective avenues for research and development."}

- {'title': 'RAGAs: Automated Evaluation of Retrieval Augmented Generation', 'year': 2023, 'summary': 'We introduce RAGAs (Retrieval Augmented Generation Assessment), a framework for reference-free evaluation of Retrieval Augmented Generation (RAG) pipelines. RAGAs is available at [https://github.com/explodinggradients/ragas]. RAG systems are composed of a retrieval and an LLM based generation module. They provide LLMs with knowledge from a reference textual database, enabling them to act as a natural language layer between a user and textual databases, thus reducing the risk of hallucinations. Evaluating RAG architectures is challenging due to several dimensions to consider: the ability of the retrieval system to identify relevant and focused context passages, the ability of the LLM to exploit such passages faithfully, and the quality of the generation itself. With RAGAs, we introduce a suite of metrics that can evaluate these different dimensions without relying on ground truth human annotations. We posit that such a framework can contribute crucially to faster evaluation cycles of RAG architectures, which is especially important given the fast adoption of LLMs.'}

- {'title': 'Benchmarking Retrieval-Augmented Generation for Medicine', 'year': 2024, 'summary': 'While large language models (LLMs) have achieved state-of-the-art performance on a wide range of medical question answering (QA) tasks, they still face challenges with hallucinations and outdated knowledge. Retrieval-augmented generation (RAG) is a promising solution and has been widely adopted. However, a RAG system can involve multiple flexible components, and there is a lack of best practices regarding the optimal RAG setting for various medical purposes. To systematically evaluate such systems, we propose the Medical Information Retrieval-Augmented Generation Evaluation (MIRAGE), a first-of-its-kind benchmark including 7,663 questions from five medical QA datasets. Using MIRAGE, we conducted large-scale experiments with over 1.8 trillion prompt tokens on 41 combinations of different corpora, retrievers, and backbone LLMs through the MedRAG toolkit introduced in this work. Overall, MedRAG improves the accuracy of six different LLMs by up to 18% over chain-of-thought prompting, elevating the performance of GPT-3.5 and Mixtral to GPT-4-level. Our results show that the combination of various medical corpora and retrievers achieves the best performance. In addition, we discovered a log-linear scaling property and the"lost-in-the-middle"effects in medical RAG. We believe our comprehensive evaluations can serve as practical guidelines for implementing RAG systems for medicine.'}

- {'title': 'LightRAG: Simple and Fast Retrieval-Augmented Generation', 'year': 2024, 'summary': 'Retrieval-Augmented Generation (RAG) systems enhance large language models (LLMs) by integrating external knowledge sources, enabling more accurate and contextually relevant responses tailored to user needs. However, existing RAG systems have significant limitations, including reliance on flat data representations and inadequate contextual awareness, which can lead to fragmented answers that fail to capture complex inter-dependencies. To address these challenges, we propose LightRAG, which incorporates graph structures into text indexing and retrieval processes. This innovative framework employs a dual-level retrieval system that enhances comprehensive information retrieval from both low-level and high-level knowledge discovery. Additionally, the integration of graph structures with vector representations facilitates efficient retrieval of related entities and their relationships, significantly improving response times while maintaining contextual relevance. This capability is further enhanced by an incremental update algorithm that ensures the timely integration of new data, allowing the system to remain effective and responsive in rapidly changing data environments. Extensive experimental validation demonstrates considerable improvements in retrieval accuracy and efficiency compared to existing approaches. We have made our LightRAG open-source and available at the link: https://github.com/HKUDS/LightRAG'}

Research Trends

- accurate
- achieve

- achieved
- answering
- assessment
- augmented
- available
- capability
- challenge
- contextually
- downstream
- enabling
- encounter
- enhance
- evaluation
- explodinggradient
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- question
- ragas
- range
- reference-free
- retrieval
- retrieval-augmented
- showcase
- shown
- source
- state-of-the-art
- store
- system
- tasks
- they
- untraceable
- when
- while
- wide

Research Gaps

None

Future Scope

None

Conclusion: Overall, the reviewed studies demonstrate significant progress in this research area while highlighting several opportunities for future work and further experimentation.

Methodology Comparison

Total Papers: 5

Comparison

- {'title': 'Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks', 'research_area': 'Large Language Models', 'research_problem': 'Large pre-trained language models have been shown to store factual knowledge in their parameters, and achieve state-of-the-art results when fine-tuned on downstream NLP tasks', 'methodology': ['Architecture', 'Model', 'Rag'], 'keywords': ['large', 'pre-trained', 'language', 'model', 'shown', 'store', 'factual', 'knowledge', 'parameter', 'achieve', 'state-of-the-art', 'when', 'fine-tuned', 'downstream', 'tasks'], 'year': 2020, 'citation_count': 17437, 'quality_score': 80, 'quality_classification': 'Very Good'}
- {'title': 'Retrieval-Augmented Generation for Large Language Models: A Survey', 'research_area': 'Large Language Models', 'research_problem': 'Large Language Models (LLMs) showcase impressive capabilities but encounter challenges like hallucination, outdated knowledge, and

non-transparent, untraceable reasoning processes', 'methodology': ['Framework', 'Model', 'System', 'Llm', 'Rag'], 'keywords': ['large', 'language', 'model', 'llms', 'showcase', 'impressive', 'capability', 'encounter', 'challenge', 'like', 'hallucination', 'outdated', 'knowledge', 'non-transparent', 'untraceable'], 'year': 2023, 'citation_count': 3936, 'quality_score': 90, 'quality_classification': 'Excellent'}

- {'title': 'RAGAs: Automated Evaluation of Retrieval Augmented Generation', 'research_area': 'General Artificial Intelligence', 'research_problem': 'We introduce RAGAs (Retrieval Augmented Generation Assessment), a framework for reference-free evaluation of Retrieval Augmented Generation (RAG) pipelines', 'methodology': ['Framework', 'Architecture', 'System', 'Llm', 'Rag'], 'keywords': ['introduce', 'ragas', 'retrieval', 'augmented', 'generation', 'assessment', 'framework', 'reference-free', 'evaluation', 'pipeline', 'available', 'https', 'github', 'explodinggradient', 'system'], 'year': 2023, 'citation_count': 958, 'quality_score': 90, 'quality_classification': 'Excellent'}
- {'title': 'Benchmarking Retrieval-Augmented Generation for Medicine', 'research_area': 'Large Language Models', 'research_problem': 'While large language models (LLMs) have achieved state-of-the-art performance on a wide range of medical question answering (QA) tasks, they still face challenges with hallucinations and outdated knowledge', 'methodology': ['Model', 'System', 'Llm', 'Rag'], 'keywords': ['while', 'large', 'language', 'model', 'llms', 'achieved', 'state-of-the-art', 'performance', 'wide', 'range', 'medical', 'question', 'answering', 'tasks', 'they'], 'year': 2024, 'citation_count': 600, 'quality_score': 90, 'quality_classification': 'Excellent'}
- {'title': 'LightRAG: Simple and Fast Retrieval-Augmented Generation', 'research_area': 'Large Language Models', 'research_problem': 'Retrieval-Augmented Generation (RAG) systems enhance large language models (LLMs) by integrating external knowledge sources, enabling more accurate and contextually relevant responses tailored to user needs', 'methodology': ['Framework', 'Model', 'System', 'Approach', 'Algorithm', 'Llm', 'Rag'], 'keywords': ['retrieval-augmented', 'generation', 'system', 'enhance', 'large', 'language', 'model', 'llms', 'integrating', 'external', 'knowledge', 'source', 'enabling', 'accurate', 'contextually'], 'year': 2024, 'citation_count': 430, 'quality_score': 80, 'quality_classification': 'Very Good'}

Highest Cited Paper

title: Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

authors: ['Patrick Lewis', 'Ethan Perez', 'Aleksandara Piktus', 'F. Petroni', 'Vladimir Karpukhin', 'Naman Goyal', 'Heinrich Kuttler', 'M. Lewis', 'Wen-tau Yih', 'Tim Rocktäschel', 'Sebastian Riedel', 'Douwe Kiela']

year: 2020

citation_count: 17437

summary: Large pre-trained language models have been shown to store factual knowledge in their parameters, and achieve state-of-the-art results when fine-tuned on downstream NLP tasks. However, their ability to access and precisely manipulate knowledge is still limited, and hence on knowledge-intensive tasks, their performance lags behind task-specific architectures. Additionally, providing provenance for their decisions and updating their world knowledge remain open research problems. Pre-trained models with a differentiable access mechanism to explicit non-parametric memory can overcome this issue, but have so far been only investigated for extractive downstream tasks. We explore a general-purpose fine-tuning recipe for retrieval-augmented generation (RAG) -- models which combine pre-trained parametric and non-parametric memory for language generation. We introduce RAG models where the parametric memory is a pre-trained seq2seq model and the non-parametric memory is a dense vector index of Wikipedia, accessed with a pre-trained neural retriever. We compare two RAG formulations, one which conditions on the same retrieved passages across the whole generated sequence, the other can use different passages per token. We fine-tune and evaluate our models on a wide range of knowledge-intensive NLP tasks and set the state-of-the-art on three open domain QA tasks, outperforming parametric seq2seq models and task-specific retrieve-and-extract architectures. For language research_problem: Large pre-trained language models have been shown to store factual knowledge in their parameters, and achieve state-of-the-art results when fine-tuned on downstream NLP tasks

methodology: ['Architecture', 'Model', 'Rag']

key_contributions: ['We introduce RAG models where the parametric memory is a pre-trained seq2seq model and the non-parametric memory is a dense vector index of Wikipedia, accessed with a pre-trained neural retriever']

future_work: []

keywords: ['large', 'pre-trained', 'language', 'model', 'shown', 'store', 'factual', 'knowledge', 'parameter', 'achieve', 'state-of-the-art', 'when', 'fine-tuned', 'downstream', 'tasks']

research_area: Large Language Models

paper_score: 80

paper_quality: Very Good

Latest Paper

title: Benchmarking Retrieval-Augmented Generation for Medicine

authors: ['Guangzhi Xiong', 'Qiao Jin', 'Zhiyong Lu', 'Aidong Zhang']

year: 2024

citation_count: 600

summary: While large language models (LLMs) have achieved state-of-the-art performance on a wide range of medical question answering (QA) tasks, they still face challenges with hallucinations and outdated knowledge. Retrieval-augmented generation (RAG) is a promising solution and has been widely adopted. However, a RAG system can involve multiple flexible components, and there is a lack of best practices regarding the optimal RAG setting for various medical purposes. To systematically evaluate such systems, we propose the Medical Information Retrieval-Augmented Generation Evaluation (MIRAGE), a first-of-its-kind benchmark including 7,663 questions from five medical QA datasets. Using MIRAGE, we conducted large-scale experiments with over 1.8 trillion prompt tokens on 41 combinations of different corpora, retrievers, and backbone LLMs through the MedRAG toolkit introduced in this work. Overall, MedRAG improves the accuracy of six different LLMs by up to 18% over chain-of-thought prompting, elevating the performance of GPT-3.5 and Mixtral to GPT-4-level. Our results show that the combination of various medical corpora and retrievers achieves the best performance. In addition, we discovered a log-linear scaling property and the"lost-in-the-middle"effects in medical RAG. We believe our comprehensive evaluations can serve as practical guidelines for implementing RAG systems for medicine.

research_problem: While large language models (LLMs) have achieved state-of-the-art performance on a wide range of medical question answering (QA) tasks, they still face challenges with hallucinations and outdated knowledge

methodology: ['Model', 'System', 'Llm', 'Rag']

key_contributions: ['To systematically evaluate such systems, we propose the Medical Information Retrieval-Augmented Generation Evaluation (MIRAGE), a first-of-its-kind benchmark including 7,663 questions from five medical QA datasets', '8 trillion prompt tokens on 41 combinations of different corpora, retrievers, and backbone LLMs through the MedRAG toolkit introduced in this work']

future_work: []

keywords: ['while', 'large', 'language', 'model', 'llms', 'achieved', 'state-of-the-art', 'performance', 'wide', 'range', 'medical', 'question', 'answering', 'tasks', 'they']

research_area: Large Language Models

paper_score: 90

paper_quality: Excellent

Common Methodologies

- Algorithm
- Approach
- Architecture
- Framework
- Llm
- Model
- Rag
- System

Research Areas

- General Artificial Intelligence
- Large Language Models

Common Keywords

- accurate
- achieve
- achieved
- answering
- assessment
- augmented
- available
- capability
- challenge
- contextually
- downstream
- enabling
- encounter
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- showcase
- shown
- source
- state-of-the-art
- store
- system
- tasks
- they
- untraceable
- when
- while
- wide

Methodology Frequency

Rag: 5

Model: 4

System: 4

Llm: 4

Framework: 3

Architecture: 2

Approach: 1

Algorithm: 1

Most Common Methodology: Rag

Quality Distribution

Very Good: 2

Excellent: 3

Publication Years

2020: 1

2023: 2

2024: 2

Citation Statistics

highest: 17437

lowest: 430

average: 4672.2

total: 23361

Comparison Summary

total_papers: 5

most_common_methodology: Rag

research_areas: ['General Artificial Intelligence', 'Large Language Models']

quality_distribution: {'Very Good': 2, 'Excellent': 3}

citation_statistics: {'highest': 17437, 'lowest': 430, 'average': 4672.2, 'total': 23361}

highest_cited_title: Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

latest_paper_title: Benchmarking Retrieval-Augmented Generation for Medicine

Research Gap Analysis

Total Papers: 5

Research Areas

- General Artificial Intelligence
- Large Language Models

Common Keywords

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- wide

Future Work

None

Research Area Distribution

Large Language Models: 4

General Artificial Intelligence: 1

Keyword Frequency

large: 4

language: 4

model: 4

knowledge: 3

llms: 3

state-of-the-art: 2

tasks: 2

generation: 2

system: 2

pre-trained: 1

shown: 1

store: 1

factual: 1

parameter: 1

achieve: 1

when: 1

fine-tuned: 1

downstream: 1

showcase: 1

impressive: 1

capability: 1

encounter: 1

challenge: 1

like: 1

hallucination: 1

outdated: 1

non-transparent: 1

untraceable: 1

introduce: 1

ragas: 1

retrieval: 1

augmented: 1

assessment: 1

framework: 1

reference-free: 1

evaluation: 1

pipeline: 1

available: 1

https: 1

github: 1

explodinggradient: 1

while: 1

achieved: 1

performance: 1

wide: 1

range: 1

medical: 1

question: 1

answering: 1

they: 1

retrieval-augmented: 1

enhance: 1

integrating: 1

external: 1

source: 1

enabling: 1

accurate: 1

contextually: 1

Research Trends

- large
- language
- model
- knowledge
- llms
- state-of-the-art
- tasks
- generation
- system
- pre-trained
- shown
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- factual
- parameter
- achieve

Gap Categories

datasets: []

models: []

evaluation: []

applications: []

general: []

Emerging Topics

- large
- language
- model
- knowledge
- llms
- state-of-the-art
- tasks
- generation
- system

Recommendations

- Investigate unexplored research directions.

Summary

dominant_area: Large Language Models

top_trend: large

future_work_items: 0

Experiment Plan

Research Areas

- General Artificial Intelligence
- Large Language Models

Recommended Datasets

None

Baseline Models

None

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

Hardware Requirements

cpu: Intel Core i7 / AMD Ryzen 7

ram: 16 GB

gpu: NVIDIA RTX 3060 or better

storage: 100 GB SSD

Validation Strategy

• Train-Test Split

• K-Fold Cross Validation

• Hyperparameter Tuning

• Statistical Significance Testing

Experimental Workflow

• Collect Dataset

• Preprocess Data

• Train Baseline Models

• Train Proposed Model

• Evaluate Performance

• Compare Results

• Analyze Errors

• Draw Conclusions

Report Summary

Dominant Research Area: Large Language Models

Top Research Trend: large

Highest Cited Paper: Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Latest Paper: Benchmarking Retrieval-Augmented Generation for Medicine

Recommended Datasets

None

Baseline Models

None

Evaluation Metrics

• Accuracy

• Precision

• Recall

- F1-Score
- ROC-AUC

Conclusion

The report has been generated using a personalized multi-agent pipeline. The explanation style was adapted according to the Intermediate profile to improve readability and usefulness.